

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation has been maintained throughout the project, making the code clean and easy to understand.
2	Proper File Structure	Files and folders are logically organized	Yes	The project is organized into separate folders for dataset, notebooks, trained model, Flask application, templates, and static files, making navigation simple.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names such as Applicant Income, Loan Amount, Credit History, and Loan Status are used for better readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their purpose, making the project workflow easier to follow and maintain.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are included at important stages such as data preprocessing, visualization, model training, and prediction to improve understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	The project is divided into logical modules including data preprocessing, exploratory data analysis, model training, prediction, and web application development.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code has been minimized by reusing functions and maintaining a structured workflow.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation has been implemented for file loading and user inputs. Advanced exception handling can be added in future improvements.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python (Pandas)	Data cleaning and feature engineering	High

2	Data Visualization	Matplotlib & Seaborn	Exploratory Data Analysis	High
3	Machine Learning Model	Scikit-learn	Loan eligibility prediction	High
4	Flask Web Application	Flask	User interface for prediction	High
5	Trained Model	Pickle	Model deployment and prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a well-organized folder structure, making files easy to locate and manage.
Readability	5	Meaningful variable names, consistent formatting, and clear logic improve the readability of the code.
Reusability	5	Individual modules can be reused in similar machine learning classification projects with minimal changes.
Documentation / Comments	4.5	The code includes comments for major sections. Additional inline documentation could further improve maintainability.
Overall Score	4.87	The project follows good coding practices with a clean structure, reusable modules, and readable code. Minor improvements in documentation and exception handling can make it even more maintainable..